

Phase 1 Discover

DAEN 459/460 Capstone Senior Design

Alexander Abuabara

Fall 2026



ENGINEERING
TEXAS A&M UNIVERSITY

GENERATIONS

Alpha

(Born 2013-2025)



Gen Z

(Born 1997-2012)



Millennials (Gen Y)

(Born 1981-1996)



Gen X

(Born 1965-1980)



Boomers

(Born 1946-1964)



Silent

(Born 1928-1945)



Bosses are firing Gen Z grads just months after hiring them

- After two years of complaints, employers are now *acting*. Many are terminating Gen Z hires within months of onboarding.
- **Top concern?**

Bosses are firing Gen Z grads just months after hiring them

- After two years of complaints, employers are now *acting*. Many are terminating Gen Z hires within months of onboarding.
- **Top concern?**
 - **Lack of motivation and initiative**
(50% of managers cite as the primary reason for termination)
- Other major issues reported:
 - Unprofessional behavior,
 - Poor organization,
 - Weak communication skills.
- Practical workplace challenges include:
 - Frequent lateness to work and meetings,
 - Inappropriate office attire,
 - Use of unsuitable language in professional settings.

Ref.:

<https://fortune.com/article/why-are-companies-firing-gen-z-employees-workplace-bosses-workers-jobs/>

<https://www.cnbc.com/2025/12/08/how-recent-grads-are-dealing-with-the-shrinking-pool-of-entry-level-jobs.html>



- Critique your own project against **learning goals** rather than technical checklists only!
- You are not being graded on reproducing a solution but on the **quality of your thinking and decisions**!
- Peers are not parallel workers, but as part of a **shared system**. Shift the culture from individual performance to collective responsibility and make **collaboration** (not cooperation only)!
- **Metacognition** ...
why I'm thinking what I'm thinking → make the invisible visible!

Reminder (of templates and more)

- [Meeting Report with Project Sponsor.docx](#)
- [Project Charter Template.docx](#)
- [Points of Contact of the Projects](#)

Readings for Wed and Friday Lab

- The Five Dysfunctions of a Team

<https://youtu.be/z8jHFk0uLlk?si=qXPD1feeKFoWX0wP>

- Bullet Journal

<https://www.thelazygeniuscollective.com/blog/how-to-bullet-journal>

- Amish Hackers

<https://kk.org/thetechnium/amish-hackers-a/>

Project Charter

Capture Clear, Actionable Project Needs

After the 1st meeting with sponsor ...

- Draft project plan → initial charter outline
- Identify gaps in requirements, assumptions, and constraints
- Re-assign roles? (notetaker, lead, etc.)
- Estimate resources, best/educated guess
(Brainstorm, plan scenarios for time, skills, any type of cost)
- Draft a sequence of activities and a project schedule framework (set of tools, processes, and methods for developing and controlling a project's timeline)
- Get started identifying risks and planning mitigations
- Finalize first project charter draft
(next week, in parallel to the project management plan if possible)

Sponsor Meeting Tips



Be prepared

- Agenda prepared and sent ahead of time
- Questions prepared
- Team knows who will speak and on what topic
- Team members assigned to take notes
(record audio/video **only** and **only if** sponsor agrees)

Listen more than you speak

- Wait for sponsor to answer questions,
don't interrupt with additional questions
- Don't try to provide immediate solutions,
this is still a 2-semester long project!

Sponsor Meeting Tips

Ask for clarification on anything you don't understand

- Ask for meaning of any terms or acronyms the sponsor uses!
Don't assume your teammates will know!
- Don't pretend to understand something you don't.

End the meeting with plans



- **Action items** for team or sponsor with target dates.
- Next meeting date set or proposed.

Sponsor Meeting Tips

Already thinking ahead ... after the meeting:

- *What went well?*
- *What do you wish you'd done prior to the meeting that you did not?*
- *Did all the team get to introduce themselves?*
- *Were your expectations met for the next meeting?*
- *Did you identify some needed information and establish a target date?*
- *Did you set a follow-up meeting or recurring meeting?*
- *What can be improved?*



Sponsor Meeting Tips

Remain positive and upbeat throughout the project

- Not everything will go perfectly, don't expect it to and don't get down when it doesn't.
- Never talk bad about anyone; sponsor, team member, instructors (especially).



If someone needs to miss a meeting

- Make sure the sponsor knows why, explain at the start of the meeting that ... *can't make it because* ...
- Make sure to follow up with person that missed to distribute information and/or action items!

Sponsor Meeting Tips



Ensure meetings don't go longer than allotted

- Listen for clues sponsor needs to go (“well, that about wraps it up...”, “OK, if there isn't anything else...”, etc.)

Don't be afraid to call the sponsor if emails are not answered and you need answers

- Set target dates for required information and follow up,
- If you call and they don't answer, leave a voicemail.

Sponsor Meeting Tips

*You'll be surprised
by the doors this
can open...*

!

- Dress **appropriately** (*ask if in doubt*) <https://careerclaset.tamu.edu/>
- Phone on **silent mode** and simply **don't use it!!**
- Follow all **safety rules** (and instructions),
- **Ask** before taking photos or videos,
- **Document** who you meet with,
- You are representing **Texas A&M**, be a good example!



Team Dynamics

1. You are called a team for a reason

- You have the same goals
- You each have skills and abilities

2. Share the load

- No one gets a free pass
- Work independently but together



Team Dynamics

3. Recognize and acknowledge differences

- Compromise
- Use individual strengths to make the team better

4. If you have difficulty

- Try to work things out together (don't talk bad about anyone!)
- Contact instructor team for assistance only if you can't work it out





Faculty Advisor

- You will have to identify a potential faculty advisor, do it ASAP!
- For areas of research interest/expertise, consult the ISEN website, follow research tab to Research Fields.
- Your project does not need to perfectly align with a professor's research field, but they are more likely to accept if it is something that interests them.



Advisors

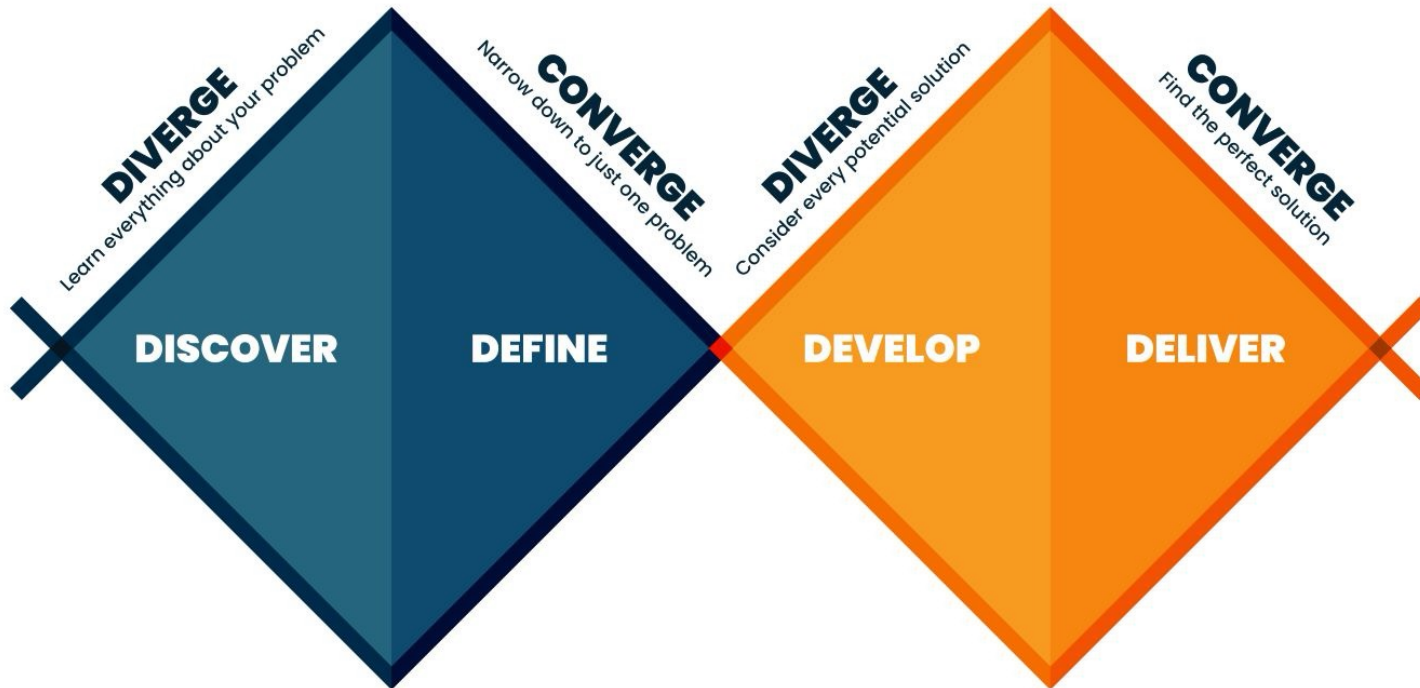
Identifying and Approaching

- Review potential matches
 - Check ISEN faculty research areas
 - Reflect on your project's needs and interests
- Reach out professionally
 - Draft a concise introductory email (project topic, goals, timeline)
 - Request a brief meeting to discuss mutual fit
- Prepare for meetings
 - Be ready to explain your project vision
 - Bring initial notes from your first meeting
- Confirm commitment
 - Secure agreement and establish possible meeting cadence
 - Inform me and Nathan once advisor is confirmed



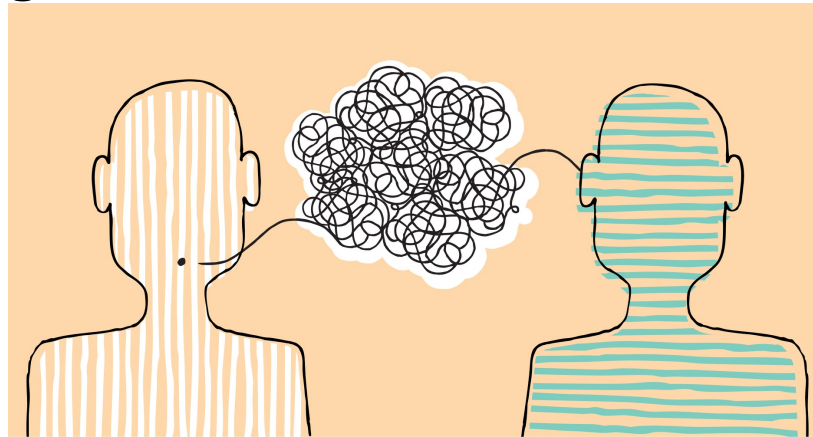
Phase 1 Discover

- Phase 1 will **set the direction** for your project.
- Phase 1 is **not about solutions**;
it is about the problem/opportunity.



Phase 1 Discover

- Phase 1 will **set the direction** for your project.
- Phase 1 is **not about solutions**;
it is about the problem/opportunity.
- Some **primary goals** for Phase 1 include
 - Information gathering,
 - Problem definition,
 - Client needs identification,
 - Technical objectives development.
- Establishing good/effective communication is also a goal!



Information Gathering

- Will occur continuously throughout the project, but critical at beginning.
- Necessary for defining parameters and scope of project.
- Information can include (but not limited to)
 - Sponsor company,
 - Industry/market,
 - Competitors,
 - **Standards/Constraints** that guide or limit the solution (ABET).
- Sources can include
 - Sponsor, trade publications or other public published material, internet.
- **The goal of Phase 1 is to gather as much information as possible,** even if you're not sure it's relevant.

Problem Definition



- Should be a **concise statement** of the problem facing your sponsor
 - Focus on the source of the problem, not the symptoms or solution,
 - Keep the language simple, avoid technical jargon,
 - Avoid detail, that will be fleshed out with customer needs.
- Done **in conjunction** with *Information Gathering*.

Problem Definition (Example)

- Should be a **concise statement** of the problem facing your sponsor
 - Focus on the source of the problem, not the symptoms or solution,
 - Keep the language simple, avoid technical jargon,
 - Avoid detail, that will be fleshed out with customer needs.
- Done **in conjunction** with Information Gathering.

Good:

The Radiology Department at Texas A&M Hospital experiences prolonged imaging turnaround times due to inefficiencies in the order processing workflow, resulting in delayed clinical decision-making.

Not so good:

The Radiology Department at Texas A&M Hospital has many problems with imaging, staff availability, communication between departments, patient flow, and rising costs, which together negatively impact hospital efficiency and patient satisfaction.

Why this is stronger?

Good:

The Radiology Department at Texas A&M Hospital experiences prolonged imaging turnaround times due to inefficiencies in the order processing workflow, resulting in delayed clinical decision-making.



Why this is stronger?

Good:

The Radiology Department at Texas A&M Hospital experiences prolonged imaging turnaround times due to inefficiencies in the order processing workflow, resulting in delayed clinical decision-making.

- **Focuses** on a single, primary process (order processing workflow)
- Names one **measurable** outcome (imaging turnaround time)
- **Avoids mixing** staffing, length of stay, and cost into the core problem
- Creates a **clear target** for data collection and pipeline design



Why this is "not so good"?

Not so good:

The Radiology Department at Texas A&M Hospital has many problems with imaging, staff availability, communication between departments, patient flow, and rising costs, which together negatively impact hospital efficiency and patient satisfaction.



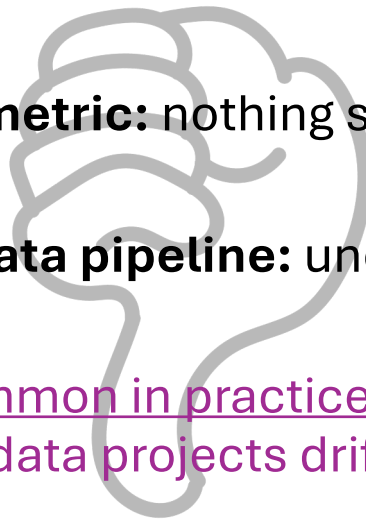
Why this is "not so good"?

Not so good:

The Radiology Department at Texas A&M Hospital has many problems with imaging, staff availability, communication between departments, patient flow, and rising costs, which together negatively impact hospital efficiency and patient satisfaction.

- **Too broad:** bundles multiple domains (staffing, communication, flow, cost, satisfaction)
- **No clear process focus:** unclear what system or workflow should be instrumented
- **No measurable primary metric:** nothing specific to optimize or monitor
- **Hard to translate into a data pipeline:** unclear what data to collect, where, or why

This kind of statement is common in practice.
And it's exactly what makes data projects drift or fail.

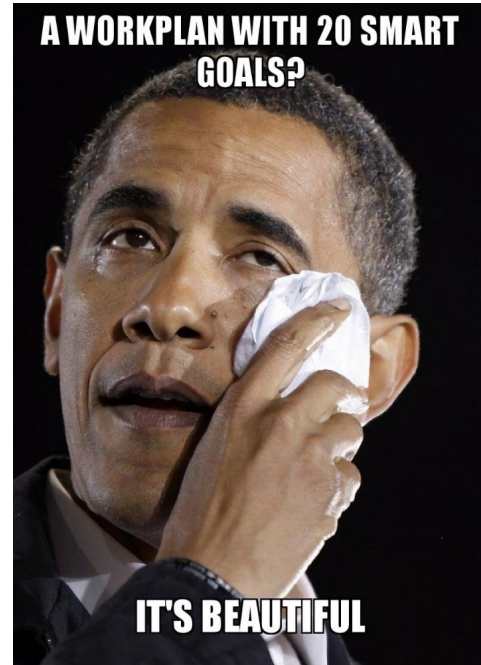


Client Needs

- Likely discovered as part of problem definition development, but this allows you to expand on the problem definition with **statements of specific needs**
- Should be a **list of things** in clear, unambiguous, non-technical terms
- May be **provided directly by the sponsor** or you may need to solicit and/or **interpret them from conversations** and other research
- Number client needs for identification in **traceability** matrix
- The client needs...
 - **N1**: Awareness of data quality, security, and compliance risks
 - **N2**: Limited exposure to corrupted, incomplete, or inconsistent data in production systems
 - **N3**: A standardized process to increase adoption and correct usage of data pipelines and tools
 - **N4**: Fewer system and pipeline failures, reducing operational downtime

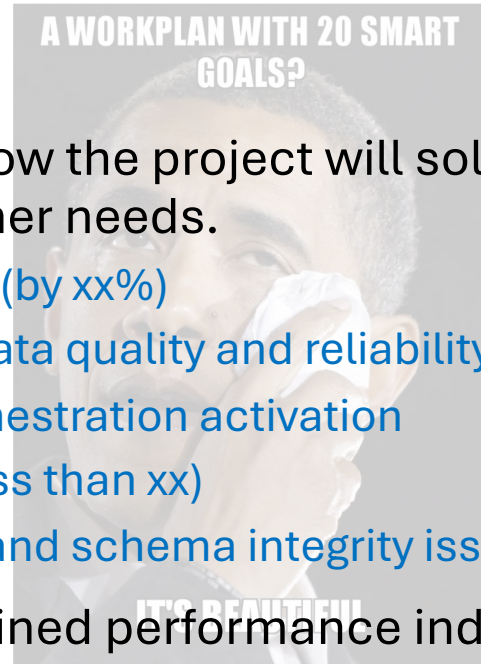
Technical Objectives

- Technical objectives should translate the customer needs and problem definition into actionable, measurable goals.
- Each objective should be **Specific, Measurable, Achievable, Relevant, and Time-bound** (SMART).



Technical Objectives

- Technical objectives should translate the customer needs and problem definition into actionable, measurable goals.
- Each objective should be **Specific, Measurable, Achievable, Relevant, and Time-bound (SMART)**.
- Number TO's for **traceability** purposes.
- Technical objectives **should address** how the project will solve the defined problem and satisfy the customer needs.
 - **TO1:** Reduce data pipeline failure rates (by xx%)
 - **TO2:** Increase engineer awareness of data quality and reliability risks
 - **TO3:** Improve automated workflow orchestration activation
 - **TO4:** Decrease system downtime (to less than xx)
 - **TO5:** Eliminate identified data leakage and schema integrity issues
- At this point your TO's may not have defined performance indicators, but **they will evolve as the project moves forward.**



Technical Objectives for Open-ended Research Projects

Your goals should center around developing **knowledge, frameworks, and methodologies** that help answer key questions or inform **decision-making**.

- If your project is more research-focused, your objectives **may not have** quantifiable performance improvement criteria.
- For research-driven projects, technical objectives should:
 - **Define Key Questions**
What are you trying to **discover, validate, or understand**?
 - **Establish Analytical Frameworks**
What **methods, models, or simulations** will you develop?
 - **Characterize System Relationships**
How do the **key factors** of your problem/system interact?
 - **Develop Conceptual or Qualitative Deliverables**
What **frameworks, decision-support tools, methodologies, or documentation** will your team generate?

See the document posted on Canvas for examples.

Example

Research-Oriented Data Engineering Project Objectives

Suppose your team is studying data pipeline reliability in large-scale streaming systems. Your technical objectives might be:

- **Key Questions**

- **What factors** most strongly influence end-to-end latency and data loss in a real-time streaming pipeline?
- **How do different** buffering strategies affect system stability under peak load?

- **Analytical Frameworks**

- **Develop a benchmarking framework** to simulate varying data arrival rates and system loads.
- **Design experiments** using controlled workloads to compare multiple pipeline architectures.

- **System Relationships**

- **Analyze how** ingestion rate, buffer size, and processing throughput interact to produce bottlenecks.
- **Model the relationship between** fault frequency and recovery time.

- **Conceptual or Qualitative Deliverables**

- **Produce a decision framework that guides engineers** in selecting buffering and checkpointing strategies.
- **Create documentation that outlines best practices** for designing resilient streaming pipelines.

Traceability

- Traceability will ensure that your project is **on track** to meet your technical objectives.
- At this point you should connect defined client needs to technical objectives in a **traceability matrix**.
- Client needs will likely not map one-to-one to technical objectives, but **all client needs should map to at least one technical objective**.

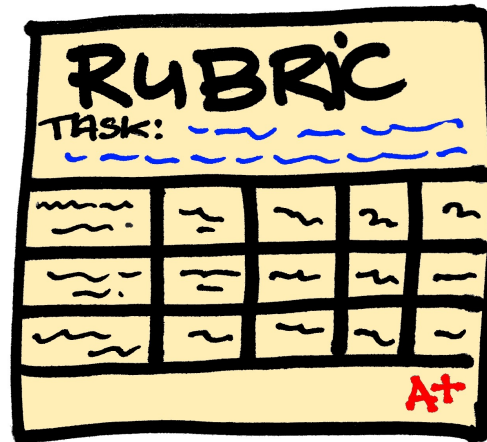
Client Needs (N)	Technical Objectives (TO)
N1	TO1, TO3, TO6
N2	TO2, TO4
...	...

Short Presentation

- Let's try to make a short presentation sometime next week.
- Do not try to present everything you've learned so far **(time is limited)**.
- Appendix is for information you've gathered that might be relevant:
 - Reference information as necessary in presentation,
 - Don't present appendix slides,
 - Technical grade for Information Gathering **will be** partially based on appendix content.
- **Practice** your presentation, both individual parts and as a team!

Phase 1 Presentation Rubric

- We all (instructor and classmates) can provide input/feedback
- Teams can practice



Drafting Questions & Follow-up Messages

- Be specific and concise
("What format should the dataset use?")
- Prioritize by urgency and impact
(critical path first)
- Avoid assumptions;
confirm facts ("*Can we assume ...?*")

Message Template

- **Greeting** – Address sponsor respectfully
- **Thank You** – Acknowledge time and input
- **Summary** – Recap key points from last meeting
- **Action Items** – List clarifications or next steps
- **Next Meeting Proposal** – Offer times/dates or request confirmation

Best Practices

- Use a single team email thread or project board
- Confirm receipt and agreements in writing
- Update faculty advisor on major changes or risks
- Always cc-me abuabara@tamu.edu

"Least But Not Last"

To keep meetings on track:

- Group leaders will also be hosts on Zoom, **and they must remove anyone who does not behave appropriately.**
- Discuss within your group what the minimum expectations should be, as this will affect everyone in the team!